

PROJECT TITLE

**AEGIS LEARN - A Generative AI Personalized Learning
Framework**

Review-1 Project Document

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Academic Year: 2026–27

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1. ABSTRACT

Academic textbooks are rich resources of foundational knowledge, but their static nature fails to accommodate the diverse learning speeds, backgrounds, and cognitive needs of individual students. Conventional educational technology often delivers uniform content without dynamic adaptation, leading to student disengagement or unresolved learning gaps. This project proposes Aegis Learn, a Generative AI-powered personalized learning framework that bridges the gap between static textbooks and tailored academic guidance. Aegis Learn integrates Retrieval Augmented Generation (RAG), custom prompt engineering, and Bloom's Taxonomy to synthesize custom educational materials and drive an adaptive learning cycle. The framework operates by ingesting corpus datasets from open-educational platforms such as OpenStax, NCERT, and Khan Academy, chunking and indexing them for semantic vector matching. During retrieval, student queries dynamically extract relevant textbook context. The system then applies a cognitive filter mapping to the six levels of Bloom's Taxonomy (from Remembering to Creating) to adjust the complexity and cognitive demand of the generated explanations, quizzes, and study guides. Furthermore, Aegis Learn incorporates a student performance loop that automatically evaluates responses, diagnoses concept weaknesses, and triggers remedial content generation to foster mastery. Evaluation of the framework demonstrates its capacity to maintain instructional alignment, suppress hallucinated information by grounding responses in verified textbook contexts, and personalize learning paths. By providing an interactive platform that includes a Bloom Content Generator, Prompt Sandbox, and Student Portal, Aegis Learn showcases the potential of modern LLMs to act as adaptive, transparent, and scalable personal tutors.

2. PROBLEM STATEMENT

Traditional e-learning platforms generally provide similar learning content to different students, despite differences in their learning speed, knowledge level, learning goals, and cognitive needs. This one-size-fits-all approach can result in learning gaps and reduced engagement. Existing Generative AI systems can generate educational content, but they may produce inaccurate or unsupported information and may not sufficiently adapt content to individual learners. There is also limited integration of Retrieval-Augmented Generation, learner personalization, Bloom's Taxonomy, and AI-based coding practice within a single Computer Science-focused framework. Therefore, there is a need for a personalized Generative AI learning framework that can retrieve reliable educational information and generate curriculum-aligned learning content according to individual student requirements.

3. OBJECTIVES

- To develop an integrated personalized Generative AI learning framework that combines student profiling, Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs) to address the limitations of existing educational AI systems.
- To generate customized study notes using Generative AI.
- To implement AI-based coding examples and technical explanations for undergraduate computer science students.
- To generate adaptive quizzes based on student learning levels.
- To integrate Bloom's Taxonomy into learning assessment.
- To optimize learning outcomes through personalized content generation and performance-based remedial learning

4. LITERATURE SURVEY

4.1 Overview

A literature survey was conducted to study existing applications of Generative Artificial Intelligence, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), automated question generation, virtual tutoring, and Bloom's Taxonomy in education. A total of 10 relevant research papers were reviewed. The studies were analyzed based on their model or approach, datasets or data sources, major outcomes, and limitations. The limitations were then compared to identify research gaps that guide the development of the proposed Aegis Learn framework.

4.2 Literature Survey					
S.No.	Research Paper	Model / Approach	Dataset / Data Used + Link	Key Outcome	Limitation
1.	<i>LearningQ: A Large-Scale Dataset for Educational Question Generation (2018)</i>	Rule-based question generation, Seq2Seq, and Attention-based Seq2Seq models.	LearningQ: 230K+ educational question pairs from TED-Ed and Khan Academy. https://github.com/AngusGLChen/LearningQ?	Existing QG models struggled with educational questions; around 30% require higher-order cognitive skills.	Focuses mainly on question generation; lacks student personalization and adaptive tutoring.
2.	<i>KHANQ: A Dataset for Generating Deep Questions in Education (2022)</i>	BERTGeneration, GPT-2, BART, T5, UniLM	KHANQ: 1,034 learner-generated deep questions from Khan Academy. https://aclanthology.org/2022.coling-1.518/?	BART and T5 performed best, but remained below human performance on several measures.	Focuses on deep-question generation; lacks a personalized learning framework.
3.	<i>How Good are Modern LLMs in Generating Relevant and High-Quality Questions at Different Bloom's Skill Levels for Indian High School Social Science Curriculum? (2024)</i>	Falcon 40B, Llama 2 70B, PaLM 2, GPT-3.5, GPT-4	510 questions + expert evaluations based on Indian high-school Social Science. https://github.com/nicyscaria/AEQG_Blooms_Evaluation_LLMs?	91.56% of generated questions were judged relevant and high quality.	Focuses on high-school Social Science rather than undergraduate Computer Science.

4.	<i>EdNet: A Large-Scale Hierarchical Dataset in Education (2020)</i>	Knowledge tracing and learner-behavior modeling	EdNet: 131M+ student interactions from 784K+ students. https://github.com/riid/ednet?	Provides large-scale student interaction data for knowledge tracing and adaptive learning research.	Does not itself provide LLM-based personalized content generation or Bloom's Taxonomy assessment.
5.	<i>A Comprehensive Review on Generative AI for Education</i>	Generative AI / LLMs	Educational research literature/existing studies. This is a review paper, so there is no single experimental dataset. https://scale.stanford.edu/ai/repository/comprehensive-review-generative-ai-education?	Identified strong potential for personalized learning, adaptive learning and content generation.	Review does not implement one unified framework combining RAG + personalization + structured assessment.
6.	<i>Automatic Question-Answer Pairs Generation Using Pre-trained Large Language Models in Higher Education (2024)</i>	Pre-trained LLMs; pipeline, joint and multi-task approaches	SQuAD (Stanford Question Answering Dataset) + author-created higher-education benchmark datasets. https://rajpurkar.github.io/SQuAD-explorer/	LLMs showed potential for generating educational question-answer pairs evaluated through automated and human assessment.	QA generation and evaluation methods still require better standardization for higher education.
7.	<i>Does ChatGPT Spell the End of Automatic Question Generation Research? (2023)</i>	ChatGPT + Prompt-based Question Generation	Experimentally generated educational assessment questions, evaluated under different	ChatGPT produced fluent and coherent questions, but many	Limited generation of higher-order thinking questions.

			generation scenarios using Bloom's Taxonomy. No separate public dataset repository is specified https://ieeexplore.ieee.org/document/10464618 .	were at lower Bloom's levels.	
8.	<i>Towards Mitigating ChatGPT's Negative Impact on Education: Optimizing Question Design through Bloom's Taxonomy (2023)</i>	ChatGPT + Bloom's Taxonomy + Evolutionary Optimization	Data Structures and Representation (DSR) examination questions from UNSW Canberra. https://sl1nk.com/2ipg79b	Used Bloom's Taxonomy to guide generation of questions requiring different levels of cognitive reasoning.	Focuses mainly on question difficulty/design, not a complete personalized learning framework.
9.	<i>How Useful Are Educational Questions Generated by Large Language Models? (2023)</i>	InstructGPT + Controllable Text Generation + Bloom's Taxonomy	Educational questions from the paper's public GitHub repository https://github.com/sabinaelkins/educational_CQG ?	LLM-generated questions were evaluated by teachers and found to be generally high-quality.	Evaluated mainly for question quality and usefulness; it does not provide a complete personalized learning framework or learner-profile adaptation.
10.	<i>Learning to Reuse Distractors to Support Multiple-Choice Question Generation in Education (2024)</i>	Context-aware question and distractor representations; data-driven models	298 educational questions + 77K multilingual distractor vocabulary. https://github.com/Distractor-Generation/DG_Survey	Context-aware models consistently outperformed static feature-based approaches.	Focuses specifically on generating/reusing MCQ distractors; it does not address personalized learning, RAG-based content retrieval, or adaptive learner profiles.

5. RESEARCH GAP IDENTIFICATION

5.1 Identified Research Gaps

The literature survey shows that existing AI-based educational systems address individual aspects of learning, such as question generation, virtual tutoring, Retrieval-Augmented Generation (RAG), learner modeling, and Bloom's Taxonomy-based assessment. However, these components are generally not integrated into one unified framework for personalized undergraduate Computer Science learning.

The following research gaps were identified:

S.No	Research Gap
1.	Limited learner personalization: Existing systems provide limited adaptation based on individual student profiles, learning levels, interests, and learning goals.
2.	Limited integration of RAG with personalized learning: Existing RAG-based systems improve response reliability but generally do not combine retrieval with personalized educational content generation.
3.	Lack of integrated Bloom's Taxonomy assessment: Existing studies use Bloom's Taxonomy mainly for question generation, but it is rarely integrated with personalized learning and continuous assessment.
4.	Limited AI-generated coding practice: Existing educational systems focus mainly on textual questions and explanations, with limited integration of personalized coding examples and programming practice.
5.	Lack of curriculum-specific personalization: Existing approaches are often designed for general education or specific subjects and do not sufficiently target undergraduate Computer Science curricula.
6.	Absence of a unified framework: There is a need for a single framework combining learner profiling, RAG, LLM-based content generation, coding practice, and adaptive assessment.

5.2 Link Between Research Gaps and Proposed Work

AegisLearn addresses these research gaps by proposing an integrated personalized AI learning framework for undergraduate Computer Science students. The framework uses student profiles to personalize learning, RAG to retrieve reliable educational information, LLMs to generate study materials and coding examples, and Bloom's Taxonomy to create adaptive assessments according to different cognitive levels.

6. INNOVATION

The proposed AegisLearn framework introduces an integrated approach to personalized AI-based learning for undergraduate Computer Science students. The main innovation is the combination of learner profiling, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), coding practice, and Bloom's Taxonomy-based assessment within a single framework.

The proposed innovations are:

- **Student Profile-Based Personalization**
The framework considers the student's selected subject, topic, learning level, and learning goal to generate suitable learning content.
- **RAG-Based Reliable Content Generation**
Retrieval-Augmented Generation is used to retrieve relevant information from trusted educational sources before generating responses, helping improve the reliability of AI-generated content.
- **Personalized Study Material Generation**
The system generates customized study notes and explanations according to the student's selected learning requirements.
- **AI-Generated Coding Practice**
The framework generates programming examples and coding exercises relevant to the selected Computer Science topic and learning level.
- **Bloom's Taxonomy-Based Adaptive Assessment**
Questions are generated across different cognitive levels of Bloom's Taxonomy, allowing students to practice from basic understanding to higher-order thinking.
- **Unified Learning Framework**
Unlike approaches that focus on only tutoring, question generation, or retrieval, AegisLearn combines these components into one personalized learning framework.

7. CREATIVITY

AegisLearn introduces a learner-centered approach where students can control and customize their learning experience according to their academic requirements. The creative aspect of the framework lies in combining AI-generated learning resources with interactive and adaptive learning activities.

The main creative elements are:

- **Personalized Learning Path**
Students select their subject, topic, learning level, and learning goal, allowing the system to generate content suited to their requirements.
- **Interactive AI Learning Support**
The system provides AI-generated explanations, study notes, examples, and learning resources instead of providing the same fixed material to every student.
- **Adaptive Quiz Generation**
The system generates questions at different Bloom's Taxonomy levels, allowing students to progress from basic knowledge to higher-order thinking.
- **Integrated Coding Practice**
For Computer Science topics, the framework can generate relevant coding examples and practice questions along with theoretical explanations.
- **Student Learning Dashboard**
A dashboard can present generated learning resources, quiz activities, and student progress in one place, providing a more interactive learning experience.
- **Combination of Multiple AI Techniques**
The creative aspect is strengthened by combining RAG, LLMs, prompt engineering, personalization, and Bloom's Taxonomy rather than using a single AI technique.

8. NOVELTY

The novelty of AegisLearn lies in integrating multiple AI-based educational techniques into a single framework specifically designed for undergraduate Computer Science learning. Existing studies generally focus on individual components such as question generation, virtual tutoring, RAG, or Bloom's Taxonomy. AegisLearn combines these components with learner-level personalization and coding practice.

The novel aspects of the proposed framework are:

- **Unified Personalized AI Learning Framework**
Combines student profiling, RAG, LLM-based generation, coding practice, and adaptive assessment in one system.
- **Combination of RAG and Personalization**
Uses retrieved educational content together with the student's selected subject, topic, learning level, and learning goal to generate more relevant learning material.
- **Bloom's Taxonomy-Based Personalized Assessment**
Generates questions according to different cognitive levels while considering the learner's selected learning requirements.
- **Computer Science-Oriented Learning**
The framework is specifically designed to support undergraduate Computer Science topics, including theoretical concepts and programming practice.
- **Integrated Theory and Coding Learning**
Combines study notes and conceptual explanations with AI-generated coding examples and programming exercises.
- **End-to-End Learning Workflow**
Provides a complete workflow from student selection and content retrieval to AI-generated learning resources, assessment, and progress tracking.

9. FEASIBILITY

The proposed AegisLearn framework is feasible to develop using currently available software technologies, open-source tools, educational datasets, and Large Language Model APIs. The project can be implemented as a web-based application with separate frontend and backend components.

The feasibility of the project is considered in the following aspects:

- **Technical**

The framework can be developed using commonly available technologies such as React/Next.js for the frontend, Python or Node.js for backend services, and LLM APIs for AI-based content generation. RAG can be implemented using document retrieval and vector databases.

- **Data**

Educational resources and publicly available datasets can be used for developing and evaluating the system. Course materials can be processed and stored in a knowledge base for retrieval.

- **Computational**

The proposed system can use existing LLM APIs or available open-source models such as Llama. Therefore, developing and testing the framework does not necessarily require training a large language model from scratch.

- **Implementation**

The framework can be developed incrementally by implementing individual modules such as student profiling, document retrieval, content generation, coding practice, and adaptive assessment.

- **Economic**

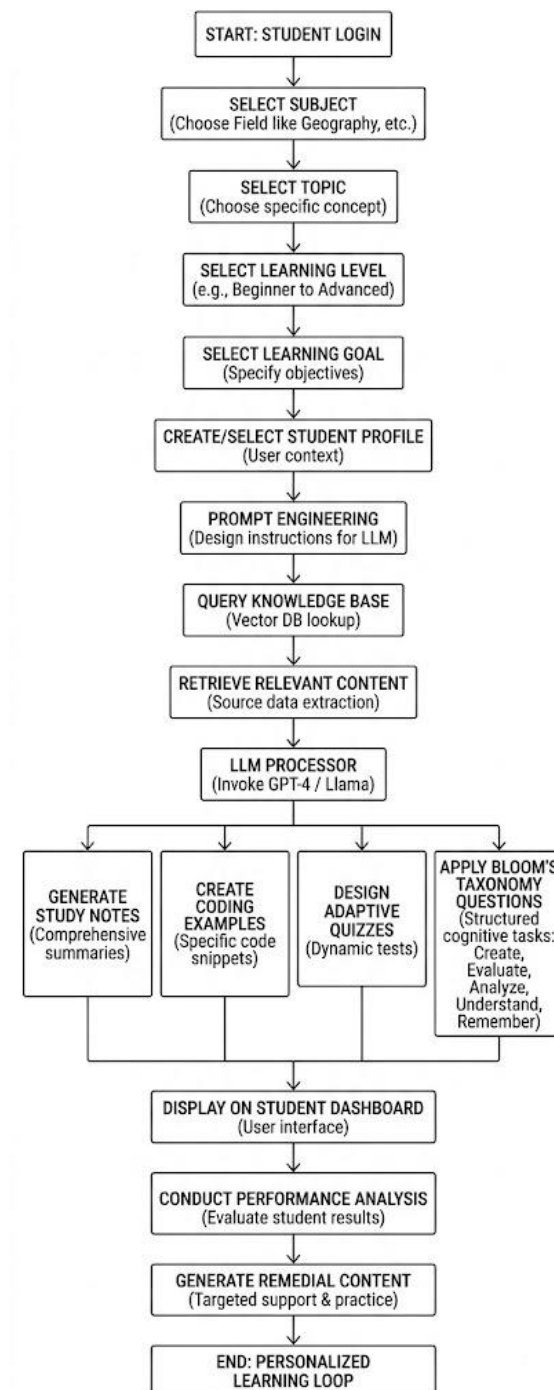
The project primarily uses open-source development tools and publicly available datasets. API usage can be controlled during development by using limited test queries and smaller models where appropriate.

- **Time**

The project can be completed in phases, beginning with literature analysis and dataset preparation, followed by RAG development, LLM integration, frontend and backend development, testing, and final evaluation.

10. PROPOSED FRAMEWORK

The proposed AegisLearn framework provides a personalized learning workflow for undergraduate Computer Science students. The framework combines student profiling, prompt engineering, Retrieval-Augmented Generation, Large Language Models, personalized content generation, coding practice, and adaptive assessment.



10.1 FRAMEWORK DESCRIPTION

The proposed framework begins by collecting the student's learning requirements, including the selected subject, topic, learning level, and learning goal. These inputs are processed through prompt engineering and passed to the Retrieval-Augmented Generation (RAG) module, which retrieves relevant information from the educational knowledge base. The retrieved information is provided to the Large Language Model (LLM) to generate personalized study materials, coding examples, and adaptive quizzes. Bloom's Taxonomy is used to structure the generated assessment questions across different cognitive levels. The student's performance can then be analyzed to provide appropriate remedial learning content.

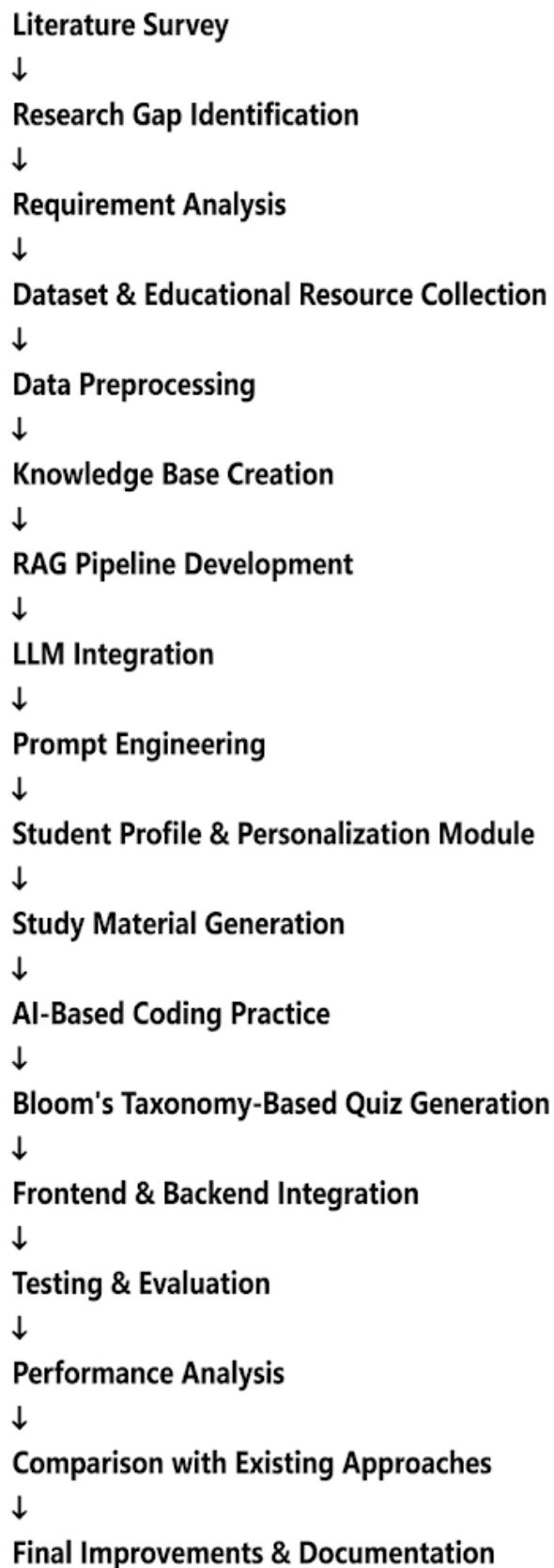
11.PROJECT PLAN

The development of AegisLearn will be carried out in multiple phases. Each phase focuses on a specific component of the proposed framework, beginning with literature analysis and data preparation and progressing toward system development, testing, and evaluation.

11.1 Project Execution Phases

Phase	Work to be Completed
Phase 1	Literature survey, identification of limitations, research gap analysis, and requirement identification
Phase 2	Collection and preprocessing of educational datasets and learning resources
Phase 3	Development of the educational knowledge base and implementation of the RAG pipeline
Phase 4	LLM integration and prompt engineering for personalized content generation
Phase 5	Development of personalized study materials, coding examples, and Bloom's Taxonomy-based adaptive quizzes
Phase 6	Frontend and backend development and integration of all modules
Phase 7	System testing, response-quality evaluation, personalization evaluation, and performance analysis
Phase 8	Final improvements, comparative analysis with existing approaches, documentation, and final presentation

11.2 Complete Workflow of Execution



11.3 Expected Evaluation

The developed framework will be evaluated based on:

- Relevance of generated educational content
- Accuracy of generated responses
- Quality of generated questions
- Alignment with Bloom's Taxonomy
- Personalization according to learner requirements
- Quality of coding examples
- Overall usefulness of the learning framework

The project will follow an iterative development process in which individual modules are developed and tested before being integrated into the complete framework. The final system will be evaluated against the identified research gaps and existing approaches to determine whether the proposed framework provides improved personalization, grounded content generation, adaptive assessment, and Computer Science-oriented learning support.

12.CONCLUSION

The literature survey indicates that Generative AI, Large Language Models, Retrieval-Augmented Generation, automated question generation, and Bloom's Taxonomy have significant potential for improving educational experiences. However, existing approaches generally focus on individual aspects such as question generation, virtual tutoring, content retrieval, or learner modeling rather than providing a unified personalized learning framework.

The proposed **AegisLearn** framework aims to address these limitations by integrating student profiling, RAG, LLM-based content generation, personalized study materials, coding practice, and Bloom's Taxonomy-based adaptive assessment into a single system for undergraduate Computer Science education.

The proposed workflow enables the system to retrieve relevant educational information, generate learning resources according to individual student requirements, provide programming practice, and assess students at different cognitive levels. The framework will be further developed and evaluated to determine its effectiveness in improving content relevance, personalization, assessment quality, and overall learning support.

13.REFERENCES

1. **Chen, G., Yang, J., Hauff, C., & Houben, G.-J. (2018).** *LearningQ: A Large-Scale Dataset for Educational Question Generation*. Proceedings of the International AAAI Conference on Web and Social Media, 12(1). DOI: 10.1609/icwsm.v12i1.14987.
2. **Gong, H., Pan, L., & Hu, H. (2022).** *KHANQ: A Dataset for Generating Deep Questions in Education*. Proceedings of the 29th International Conference on Computational Linguistics, 5925–5938.
3. **Scaria, N., Chenna, S. D., & Subramani, D. (2024).** *How Good are Modern LLMs in Generating Relevant and High-Quality Questions at Different Bloom's Skill Levels for Indian High School Social Science Curriculum?* Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications, 1–10.
4. **Choi, Y., Lee, Y., Shin, D., Cho, J., Park, S., Lee, S., Baek, J., Bae, C., Kim, B., & Heo, J. (2020).** *EdNet: A Large-Scale Hierarchical Dataset in Education*. Artificial Intelligence in Education, Lecture Notes in Computer Science, 12164, 69–73. DOI: 10.1007/978-3-030-52240-7_13.
5. **Mittal, U., Sai, S., Chamola, V., & Sangwan, D. (2024).** *A Comprehensive Review on Generative AI for Education*. IEEE.
6. **Ling, J., & Afzaal, M. (2024).** *Automatic Question-Answer Pairs Generation Using Pre-trained Large Language Models in Higher Education*. Computers and Education: Artificial Intelligence, 6, 100252. DOI: 10.1016/j.caeai.2024.100252.
7. **Does ChatGPT Spell the End of Automatic Question Generation Research? (2023).** *2023 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI)*. DOI: 10.1109/CVMI59935.2023.10464618.
8. **Elsayed, S. (2023).** *Towards Mitigating ChatGPT's Negative Impact on Education: Optimizing Question Design through Bloom's Taxonomy*. arXiv.
9. **Elkins, S., Kochmar, E., Cheung, J. C. K., & Serban, I. (2023).** *How Useful are Educational Questions Generated by Large Language Models?* arXiv:2304.06638.
10. **Bitew, S. K., Hadifar, A., Sterckx, L., Deleu, J., Develder, C., & Demeester, T. (2024).** *Learning to Reuse Distractors to Support Multiple-Choice Question Generation in Education*. IEEE Transactions on Learning Technologies, 17, 375–390. DOI: 10.1109/TLT.2022.3226523.